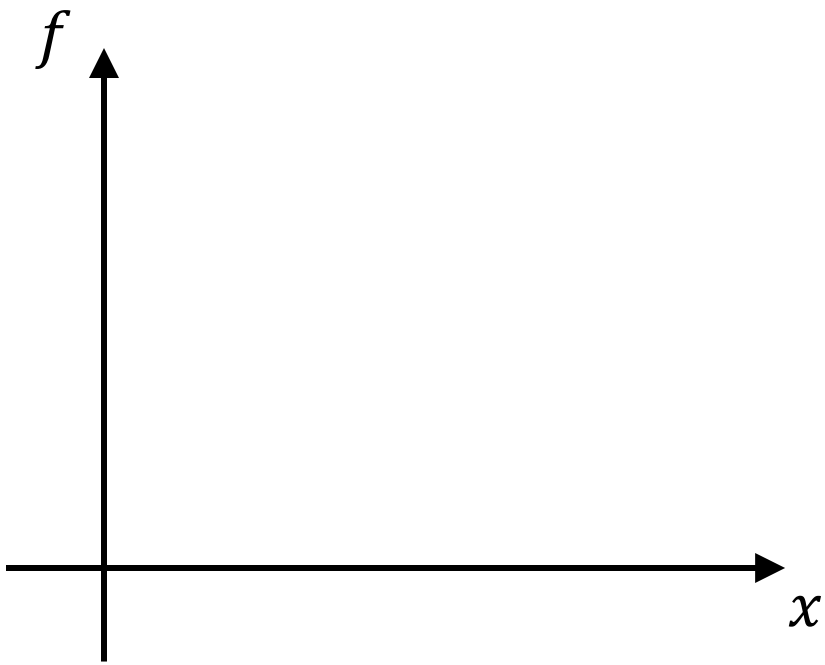
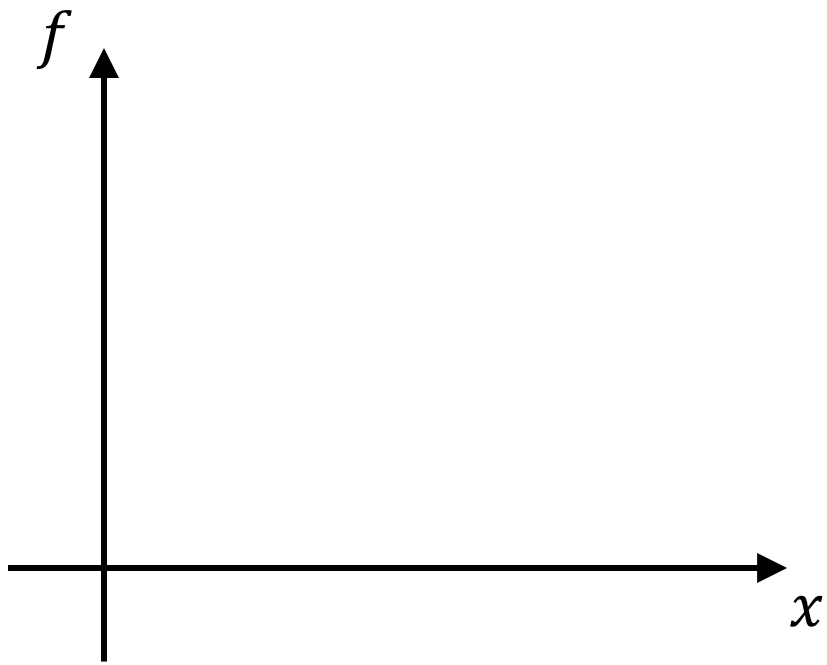
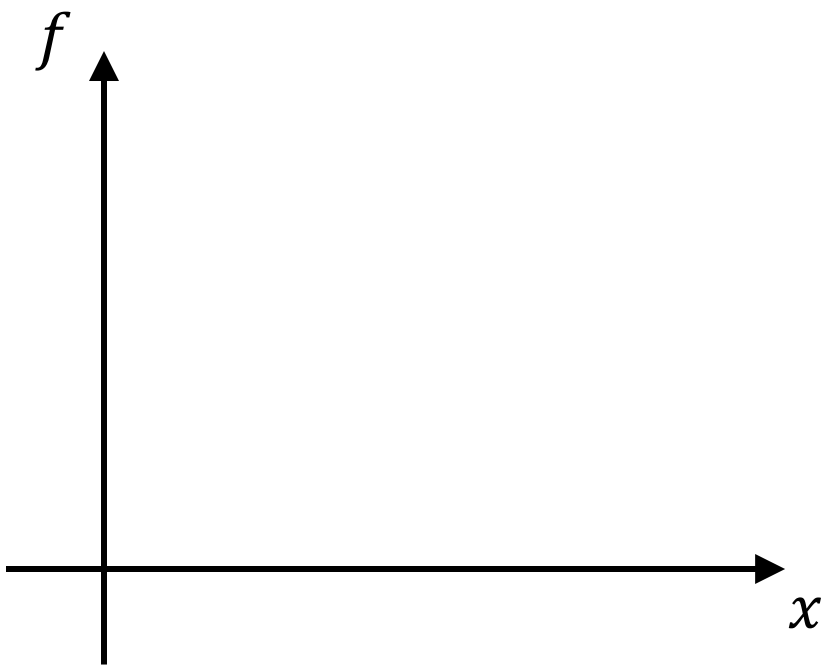
6.2 Integration

Example. Let's find the area under the graph of $f(x) = x^2$ for $3 \leq x \leq 6$.

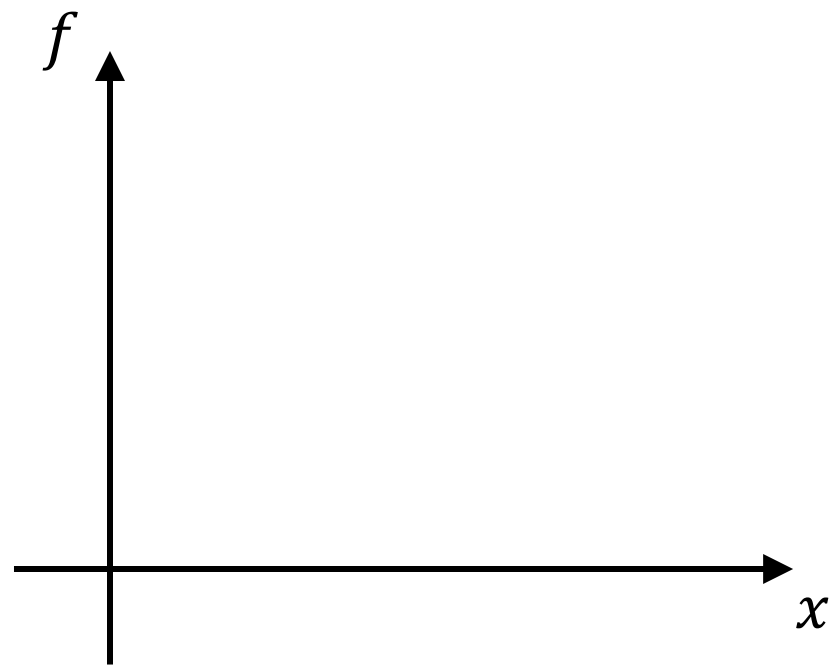




We need not have the intervals have the same length.



For general functions, max/min on each interval need not occur at the end points.



Definition 6.8. *A* <u>*partition*</u> *of* $[a, b]$ *is a set*

$$P = \{x_0, x_1, x_2, \dots, x_{n-1}, x_n\}$$

where

$$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b.$$

We refer to each interval $[x_{i-1}, x_i]$ *as a* <u>*sub-interval*</u> *of* $[a, b]$.

Example.

Definition 6.9. *Let* $f\colon [a,b] \to \mathbb{R}$ *be bounded on* $[a,b]$ *and let*

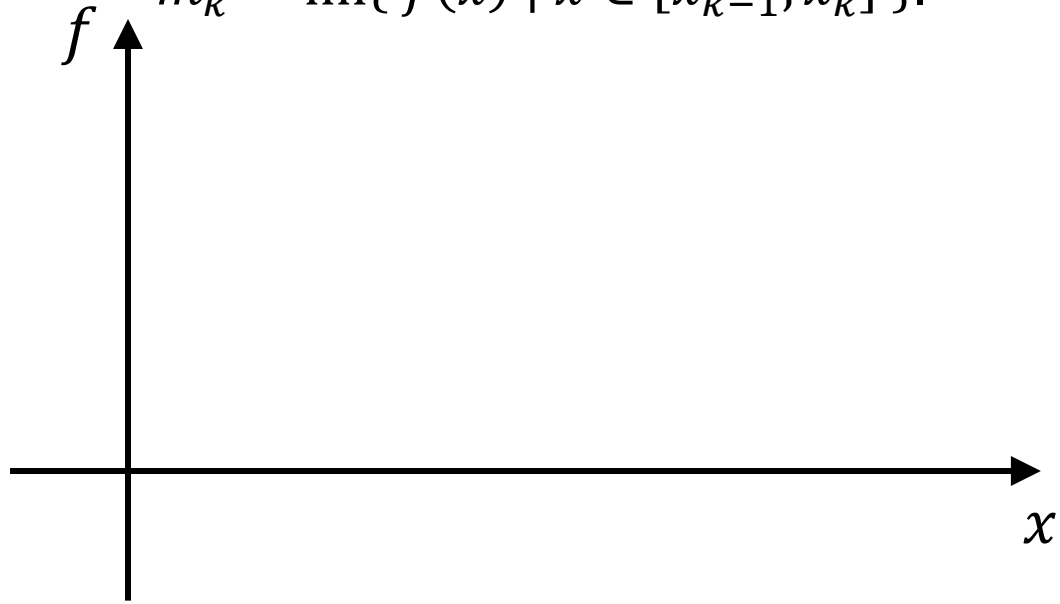
$$P = \{x_0, x_1, x_2, \dots, x_{n-1}, x_n\}$$

be a partition of $[a,b]$. *The* <u>*lower Riemann sum*</u> *of* f *with respect to* P *is given by*

$$L(f,P) = \sum_{k=1}^{n} m_k(x_k - x_{k-1}) = m_1(x_1 - x_0) + m_2(x_2 - x_1) + m_3(x_3 - x_2) + \cdots + m_n(x_n - x_{n-1})$$

where

$$m_k = \inf\{\, f(x) \mid x \in [x_{k-1}, x_k] \,\}.$$



Definition 6.9. *Let* $f\colon [a,b] \to \mathbb{R}$ *be bounded on* $[a,b]$ *and let*

$$P = \{x_0, x_1, x_2, \dots, x_{n-1}, x_n\}$$

be a partition of $[a,b]$. *The* <u>*upper Riemann sum*</u> *of* f *with respect to* P *is given by*

$$U(f,P) = \sum_{k=1}^{n} M_k (x_k - x_{k-1})$$

where

$$M_k = \sup\{\, f(x) \mid x \in [x_{k-1}, x_k] \,\}.$$

Example. Underestimate the area under $f(x) = 25 - x^2$ on $1 \leq x \leq 4$.

Notice that for each sub-interval $[x_{k-1}, x_k]$ we have $m_k \leq M_k$. Therefore

$$L(f,P) = \sum_{k=1}^{n} m_k(x_k - x_{k-1}) \leq \sum_{k=1}^{n} M_k(x_k - x_{k-1}) = U(f,P)$$

This gives us:

Lemma 6.10. *Let $f: [a,b] \to \mathbb{R}$ be bounded on $[a,b]$. For every partition P of $[a,b]$ we have*

$$L(f,P) \leq U(f,P).$$

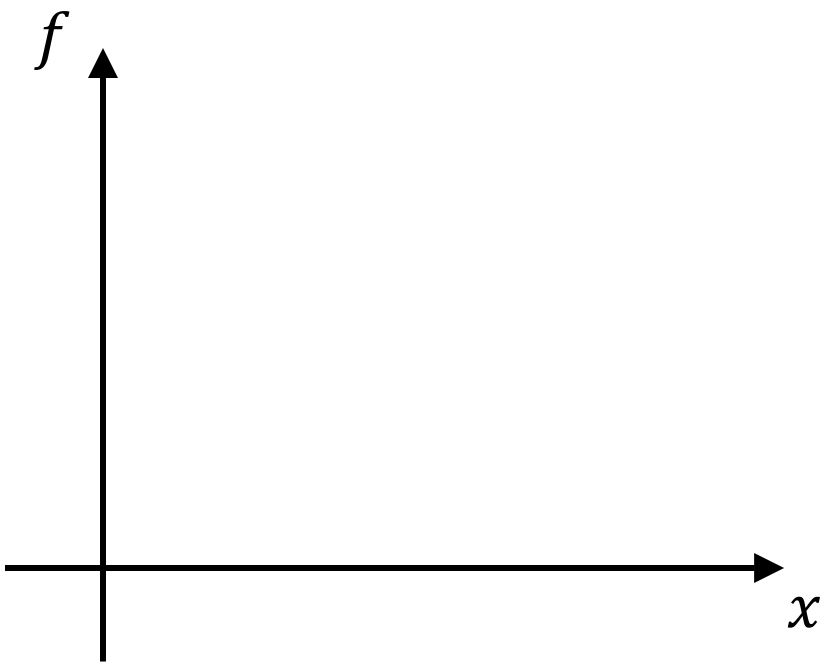
We ideally want: *Let $f: [a,b] \to \mathbb{R}$ be bounded on $[a,b]$. For every partition P, P' of $[a,b]$ we have*

$$L(f,P) \leq U(f,P').$$

Definition 6.11. *Let P and Q be partitions of $[a, b]$.*

We say that Q is a <u>refinement</u> of P when $P \subseteq Q$.

Example. Let $P = \{x_0, x_1, x_2, x_3\}$ be a partition of $[a, b]$ and Q be the refinement obtained by adding a point z between x_1 and x_2.



By using induction, we can include more points in the refinement of P, increasing the value of $L(f, P)$.

We can make the same argument for upper sums. Hence:

Lemma 6.12. *Let* $f: [a, b] \to \mathbb{R}$ *be bounded on* $[a, b]$ *and let* P *be a partition of* $[a, b]$. *If* Q *is a refinement of* P, *then*

$$L(f, P) \leq L(f, Q) \quad \text{and} \quad U(f, Q) \leq U(f, P).$$

Theorem 6.13. *Let* $f: [a, b] \to \mathbb{R}$ *be bounded on* $[a, b]$. *If* P *and* P' *be partitions of* $[a, b]$, *then*

$$L(f, P) \leq U(f, P').$$

Definition 6.14. *Let* $f\colon [a,b] \to \mathbb{R}$ *be bounded on* $[a,b]$.

The <u>*lower Riemann integral*</u> *of* f *on* $[a,b]$ *is*

$$L(f) = \sup\{\, L(f,P) \mid P \text{ is a partition of } [a,b] \,\}$$

The <u>*upper Riemann integral*</u> *of* f *on* $[a,b]$ *is*

$$U(f) = \inf\{\, U(f,P) \mid P \text{ is a partition of } [a,b] \,\}$$

Notice that for any partition P of $[a,b]$

$$L(f,P) \leq L(f) \quad \text{and} \quad U(f) \leq U(f,P).$$

As the partitions P get finer, we expect $L(f)$ and $U(f)$ to get closer together so that

$$L(f) = U(f).$$

However, this might not occur, i.e. we might have $L(f) < U(f)$.

Definition 6.15. *Let* $f\colon [a,b] \to \mathbb{R}$ *be bounded on* $[a,b]$.

We say f *is* <u>*Riemann*</u> <u>*integrable*</u> *on* $[a,b]$ *when* $L(f) = U(f)$. *In this case we denote the common value of* $L(f) = U(f)$ *by*

$$\int_a^b f(x)\,dx\,.$$

In other words:

$$L(f) = U(f) = \int_a^b f(x)\,dx\,.$$

Example. Define the *rational indicator function* $\mathfrak{I}: [1,2] \to \mathbb{R}$ by

$$\mathfrak{I}(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

Note.

1. We defined the integral of f only using concepts in real analysis, such as sup and inf, and without specifically referring to area.

   One reason is that it is hard to draw good pictures of functions which represent the behavior of the integral in full generality.

   E.g. we often draw graphs of positive continuous functions.

   Another reason is we can readily generalize integration, e.g. of $f: [a, b] \to \mathbb{R}^n$.

2. An integral is the sup/inf of a set. Computing the integral amounts to finding the value of this sup/inf.

3. The rational indicator function $\mathfrak{I}$ is not integrable on any domain $[a, b]$. And so the notation

$$\int_a^b \mathfrak{I}(x)\, dx$$

   is meaningless.

Theorem. *Let* $f: [a, b] \to \mathbb{R}$ be bounded.

f *is Riemann integrable on* $[a, b]$ *if and only if for every* $\epsilon > 0$ *there exists a partition* P_ϵ *of* $[a, b]$ such that

$$U(f, P_\epsilon) - L(f, P_\epsilon) < \epsilon.$$

Theorem 6.16. *Let* $f: [a, b] \to \mathbb{R}$. *If* f *is continuous on* $[a, b]$, *then* f *is integrable on* $[a, b]$.

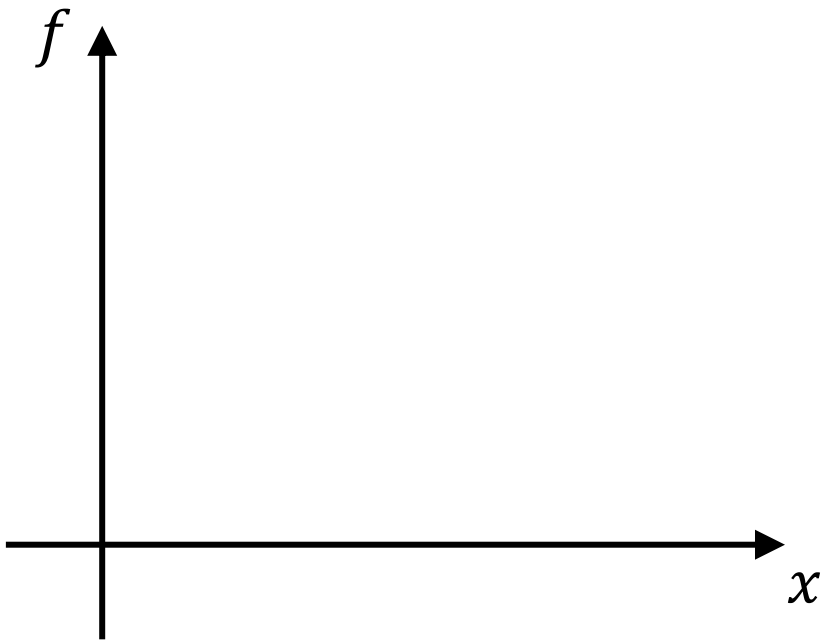
Theorem. *Let* $f: [a, b] \to \mathbb{R}$ *be integrable on* $[a, b]$.

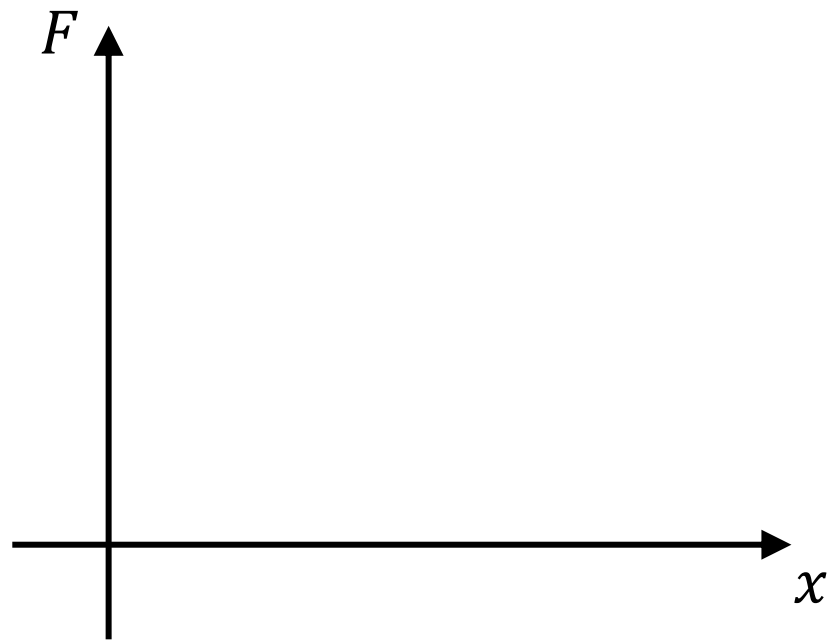
1. $\int_a^b f(x)\, dx = \int_a^c f(x)\, dx + \int_c^b f(x)\, dx$ *for all* $c \in (a, b)$.
2. $\int_a^b (f(x) + g(x))\, dx = \int_a^b f(x)\, dx + \int_a^b g(x)\, dx$
3. $\int_a^b \lambda\, f(x)\, dx = \lambda \int_a^b f(x)\, dx$ *for all constants* $\lambda \in \mathbb{R}$.
4. $\left| \int_a^b f(x)\, dx \right| \leq \int_a^b |f(x)|\, dx$
5. *If* $f(x) \leq g(x)$ *for all* $x \in [a, b]$, *then* $\int_a^b f(x)\, dx \leq \int_a^b g(x)\, dx$.

Let $f: [a, b] \to \mathbb{R}$ be integrable and $F: [a, b] \to \mathbb{R}$ be differentiable with

$$F'(x) = f(x) \text{ for all } x \in [a, b].$$

Take any partition P and consider the sub-interval $[x_2, x_3]$.





Theorem 6.18. (Fundamental Theorem of Calculus – Part I)
Let $f: [a, b] \to \mathbb{R}$ *be integrable and* $F: [a, b] \to \mathbb{R}$ *be differentiable with*

$$F'(x) = f(x) \text{ for all } x \in [a, b]$$

then

$$\int_a^b f(x)\, dx = F(b) - F(a).$$

Theorem 6.18. (Fundamental Theorem of Calculus – Part II)
Let $g: [a, b] \to \mathbb{R}$ *be integrable and* $G: [a, b] \to \mathbb{R}$ *such that*

$$G(x) = \int_a^x g(t)\, dt$$

then

i. *The function* G *is continuous on* $[a, b]$.
ii. *If* g *is continuous at* $c \in [a, b]$, *then* G *is differentiable at* c *and*

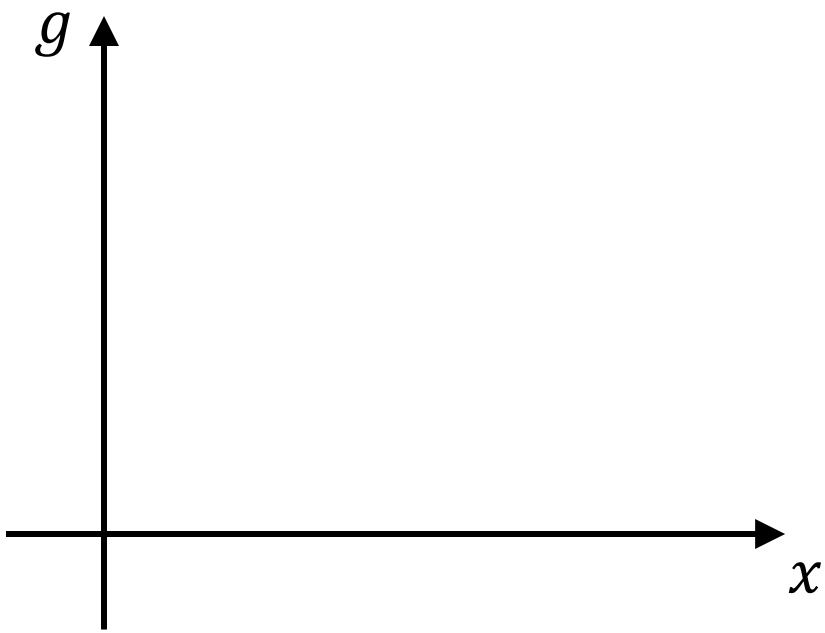
$$G'(c) = g(c).$$

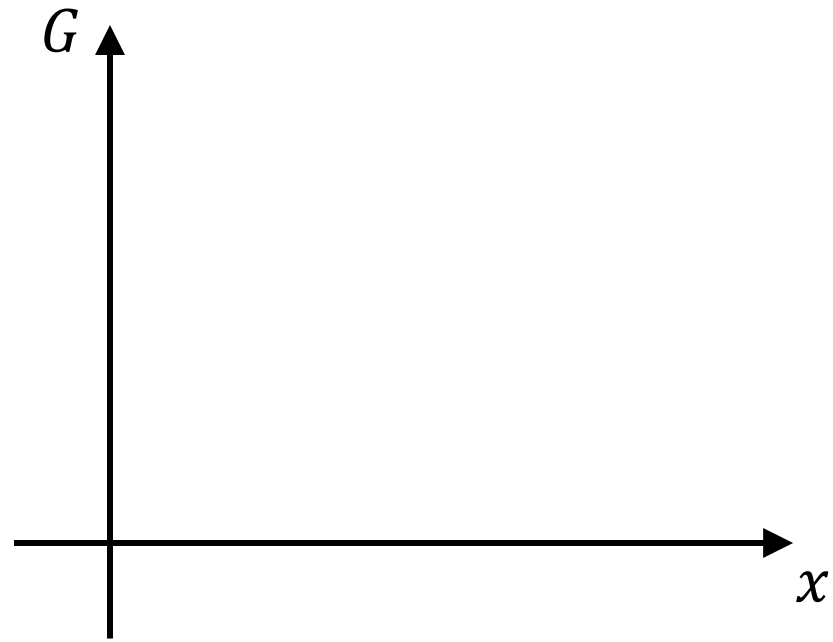
Sketch of proof of (i). Since integrable functions are bounded, there exists $M > 0$ such that

Hence for all $a \leq x < y \leq b$

Heuristic for (ii). Let g be continuous at $c \in [a, b]$. Then

Example. Let $g\colon [a,b] \to \mathbb{R}$ be the constant function $g(x) = k$.





Proof of (ii). Let $g\colon [a,b] \to \mathbb{R}$ be integrable and $G\colon [a,b] \to \mathbb{R}$ such that

$$G(x) = \int_a^x g(t)\, dt\,.$$

Let g be continuous at $c \in [a,b]$. We want to show that

$$G'(c) = g(c).$$